Master shell chronology and multi-proxy geochemical records illustrate oceanographic variability in the mid-Atlantic Region (USA) since 1800 CE

Thatcher, Diana L.[1]; Sun, Heeyeon[2]; Wanamaker, Alan D.[1]; Whitney, Nina M.[3]; Jarosinski, Lindsey[1]; Stewart, Joseph A.[4]; Williams, Branwen[5]; LaVigne, Michèle[6]; Ummenhofer, Caroline C.[7]; Remaili, Soraya S.[8]; Biastoch, Arne[9]; Martin, Torge[9]

[1] *Iowa State University, Ames, IA, USA*
[2] *University of Massachusetts Amherst, Amherst, MA, USA*
[3] *Western Washington University, Bellingham, WA, USA*
[4] *University of Bristol, Bristol, UK*
[5] *Claremont McKenna College, Claremont, ME, USA*
[6] *Bowdoin College, Brunswick, ME, USA*
[7] *Woods Hole Oceanographic Institution, Woods Hole, MA, USA*
[8] *University of South Carolina, Columbia, SC, USA*
[9] *GEOMAR Helmholtz Centre for Ocean Research, Kiel, Germany*

Instrumental data from the Atlantic Ocean along the U.S. and Canada coast indicate warming and salinity variability in recent decades, potentially associated with changes in the strength of the Atlantic Meridional Overturning Circulation (AMOC). However, observational time series of hydrographic conditions from buoys and moorings, especially at depth, are short. Additional information spanning multiple decades to centuries is needed to place recent hydrographic conditions in context and to better understand past oceanographic changes. To provide this context, we constructed a continuous master shell growth chronology spanning the last two centuries and provided geochemical records (stable oxygen and nitrogen isotopes and radiocarbon) from the Mid-Atlantic shelf region using shells of the marine bivalve Arctica islandica. Shells were collected on the outer shelf region near Ocean City, Maryland, at approximately 60 m water depth. The multiproxy records from shells from this site, located midway between the RAPID (26° N) and the OSNAP (52–60° N) instrumental arrays, provide context for present and future variability in AMOC. Output from the VIKING20X simulations (1/20° Atlantic configuration nested in ¼° global ocean, forced by the 55-year Japanese Atmospheric Reanalysis product) are used to infer the relative contributions of source water oxygen isotope (from salinity) and temperature variations on the shell oxygen isotope record since ~1970 CE, helping to understand oceanographic variability in this region. The shell oxygen isotope record ($\delta^{18}O_{shell}$) is strongly inversely correlated with the RAPID array which, although relatively short in duration, estimates AMOC strength, providing evidence that this location is sensitive to large-scale ocean dynamics.